JAYLEN WANG

Carnegie Mellon University
Department of Electrical and Computer Engineering

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BRIEF BIOGRAPHY

My work bridges computer architecture and software systems, demonstrating the importance of that bridge in enabling sustainable data center systems via solutions that span the compute stack.

As the demand for web services continues to grow, data centers are scaling up to meet the demand, consuming a massive amount of energy and producing significant carbon emissions. **My research addresses the growing carbon emissions of data centers by analyzing inefficiencies across computer architecture and software systems and designing solutions to make these systems more energy and carbon efficient.**

My work is one of the first to examine the environmental impact of hyperscale web systems and to provide actionable insights to reduce it. My work integrates carbon efficiency into computer system design, as it is crucial for sustainable growth and access to critical web services in both developed and developing nations. Critically, my carbon-efficient solutions are starting to be deployed at hyperscale and are estimated to reduce global carbon emissions by ∼0.2% by 2030—equivalent to eliminating the annual emissions of entire countries like Qatar/Venezuela.

My research has been recognized with an IEEE Micro Top Pick, the 2026 IEEE Computer Society TCuArch Graduate Student Industry Impact Award, an NSF Graduate Research Fellowship Program (GRFP) Award, the 2023 Benjamin Garver Lamme/Westinghouse Graduate Fellowship, a 2023 Ford Foundation Predoctoral Fellowship Honorable Mention, and the Jack and Mildred Bowers Scholarship in Engineering.

EDUCATION

Ph.D., Electrical and Computer Engineering — *Carnegie Mellon University*
Advisor: Prof. Akshitha Sriraman — *Aug 2022 – Present*
NSF GRFP Fellow
Dissertation Title: Enabling Sustainable Data Centers

B.Sc., Electrical Engineering — *Harvard University*
Research advisors: Profs. David Brooks & Gu-Yeon Wei — *Aug 2018 – May 2022*
Minor in Computer Science
Graduated *Summa Cum Laude* (GPA: 4.0/4.0); *Member of Phi Beta Kappa*

AWARDS AND HONORS

IEEE Computer Society TCuArch Graduate Student Industry Impact Award — *2026*
One of two recipients, for technical contributions to *GreenSKU* during 2023 Microsoft internship

***GreenSKU* selected as an IEEE Micro Top Pick** — *2025*
Awarded to the top 12 computer architecture papers of 2024

Finalist (2nd place) at SOSP Student Research Competition (SRC), Graduate Category — *2024*
At SOSP'24 SRC, selected as finalist after poster session, and 2nd place after presenting

Jack and Mildred Bowers Scholarship in Engineering — *2024*
Full tuition support for third year of PhD

NSF Graduate Research Fellowship Program (GRFP) Award Winner — *2023*
Awarded $171,000 as part of prestigious fellowship supporting exceptional graduate students in STEM

Ford Foundation Predoctoral Fellowship Honorable Mention — *2023*
Honorable mention given to top Ford Fellowship candidates

Benjamin Garver Lamme/Westinghouse Graduate Fellowship — *2023*
Full tuition support for second year of PhD

Carnegie Institute of Technology Dean's Fellow — *2022*
Awarded $83,000 towards tuition, stipend, and travel

Harvard SEAS Dean's Engineering Design Award *2022*
Awarded $500 for having one of the top 7 (out of 43) best Senior engineering design projects

Sophia Freund Prize *2022*
Awarded $1000 as highest ranking undergraduate in the Electrical Engineering department

Phi Beta Kappa Member *2022*
Admitted into Harvard's chapter, one of 146 out of 1962 (7.4%) students

Derek Bok Center Distinction in Teaching *2020, 2021*
Awarded to most highly rated (by students) TAs; received distinction in three semesters

John Harvard Scholar *2020*
Award given to top 5% (4.0 GPA) of students in respective class

Harvard College Research Program Funding Recipient *2019*
Awarded $3,500 to perform independent research during the summer

Detur Prize Winner *2019*
Recognizes students with top academic standing in their first year at Harvard

PEER-REVIEWED CONFERENCE AND JOURNAL PUBLICATIONS

- **Jaylen Wang**, Melissa Z. Pan, Udit Gupta, and Akshitha Sriraman. *(Under submission) Title omitted for anonymity.* International Symposium on High-Performance Computer Architecture (**HPCA 2027**).
- **Jaylen Wang**, Daniel S. Berger, Fiodar Kazhamiaka, Celine Irvene, Chaojie Zhang, Esha Choukse, Kali Frost, Rodrigo Fonseca, Brijesh Warrier, Chetan Bansal, Jonathan Stern, Ricardo Bianchini, and Akshitha Sriraman. *Enabling Sustainable Cloud Computing with Low-Carbon Server Design.* IEEE Micro. Issue: **Top Picks** in Computer Architecture from Conferences in 2024. Jun 2025.

  Shows how, for the first time, to design and build real production-ready, low-carbon servers and how to evaluate servers' carbon-saving potential at scale prior to deployment.
- **Jaylen Wang**, Daniel S. Berger, Fiodar Kazhamiaka, Celine Irvene, Chaojie Zhang, Esha Choukse, Kali Frost, Rodrigo Fonseca, Brijesh Warrier, Chetan Bansal, Jonathan Stern, Ricardo Bianchini, and Akshitha Sriraman. *Designing Cloud Servers for Lower Carbon.* 51st International Symposium on Computer Architecture (**ISCA 2024**). Jun 2024.
  IEEE Micro Top Picks
  Received the Artifact "Available", "Functional", and "Reproducible" ACM badges

  First work to systematically show how to design cloud servers to significantly reduce cloud carbon emissions while meeting performance goals; this solution is being adopted across Microsoft Azure, with estimated savings of ∼0.2% of global carbon emissions—on par with eliminating Qatar/Venezuela's annual emissions

PEER-REVIEWED WORKSHOP PUBLICATIONS & POSTERS

- **Jaylen Wang**, Fiodar Kazhamiaka, Pulkit Misra, Daniel S. Berger, and Akshitha Sriraman. *Safe CPU Oversubscription: One Size Does Not Fit All.* 3rd Workshop on Hot Topics in System Infrastructure (**HotInfra 2026**) held in conjunction with **ISCA 2026**. Jun 2026.
- Abnash Bassi, **Jaylen Wang**, Fiodar Kazhamiaka, Daniel S. Berger, and Akshitha Sriraman. *Store or Recompute? Characterizing the Carbon Tradeoff of KV Cache Retention in LLM Inference.* 11th Workshop on Energy Efficient Machine Learning and Cognitive Computing (**EMC2 2026**). Mar 2026.
- **Jaylen Wang**, Melissa Pan, Udit Gupta, and Akshitha Sriraman. *Giving Old Servers New Life at Hyperscale.* Presented Poster at **SOSP 2024**. Nov 2024.
- **Jaylen Wang**, Asser Tantawi, Olivier Tardieu, and Akshitha Sriraman. *Making Multi-Cluster Scheduling Carbon-Aware.* 2nd Doctoral Workshop (**SysDW 2024**) held in conjunction with **SOSP 2024**. Nov 2024.
- **Jaylen Wang**, Daniel S. Berger, Fiodar Kazhamiaka, Celine Irvene, Chaojie Zhang, Esha Choukse, Kali Frost, Rodrigo Fonseca, Brijesh Warrier, Chetan Bansal, Jonathan Stern, Ricardo Bianchini, and Akshitha Sriraman. *Designing Cloud Servers for Lower Carbon.* 2nd Workshop on Hot Topics in System Infrastructure (**HotInfra 2024**) held in conjunction with **SOSP**. Nov 2024.

– **Jaylen Wang**, Melissa Pan, Udit Gupta, and Akshitha Sriraman. *Giving Old Servers New Life at Hyperscale.* 6th Young Architect Workshop (**YArch 2024**) held in conjunction with **ASPLOS**. Jun 2024.

– **Jaylen Wang**, Udit Gupta, and Akshitha Sriraman. *Peeling Back the Carbon Curtain: Carbon Optimization Challenges in Cloud Computing.* 2nd Workshop on Sustainable Computer Systems (**HotCarbon 2023**). Jul 2023.

Performs the first ever characterization of server generations for microservice-based web services to enable hardware lifetime extension

– Jialun Lyu, **Jaylen Wang**, Kali Frost, Chaojie Zhang, Celine Irvene, Esha Choukse, Rodrigo Fonseca, Ricardo Bianchini, Fiodar Kazhamiaka, and Daniel S. Berger. *Myths and Misconceptions Around Reducing Embedded Carbon for Cloud Platforms.* 2nd Workshop on Sustainable Computer Systems (**HotCarbon 2023**). Jul 2023.

– **Jaylen Wang**, Udit Gupta, and Akshitha Sriraman. *Giving Old Servers New Life at Hyperscale.* 1st Workshop on Hot Topics in System Infrastructure (**HotInfra 2023**) held in conjunction with **ISCA**. Jun 2023.

– **Jaylen Wang**, Udit Gupta, and Akshitha Sriraman. *Characterizing Datacenter Server Generations for Lifetime Extension and Carbon Reduction.* 1st Workshop on NetZero Carbon Computing (**NetZero 2023**) held in conjunction with **HPCA**. Feb 2023.

– Sahana Rangarajan, **Jaylen Wang**, Sara Mahdizadeh Shahri, Pratyush Patel, and Akshitha Sriraman. *Designing Equitable Data Center Scheduling Systems.* Career Workshop for Inclusion and Diversity in Computer Architecture (**CWIDCA 2022**) held in conjunction with **MICRO**. Oct 2022.

Introduces equity as a first-order design metric in modern data center scheduling systems

– **Jaylen Wang**, Abdulrahman Mahmoud, Gu-Yeon Wei, and David Brooks. *A Dataflow-Aware Fault Resilience Analysis Framework for Deep Neural Network Accelerators.* 4th Young Architect Workshop (**YArch 2021**) held in conjunction with **ASPLOS**. Mar 2022.

Introduces a new framework and tool to quickly and accurately assess the reliability of deep neural network accelerator designs to random bit flips, providing insights for resilient accelerator design

INVITED SEMINAR TALKS

Designing Cloud Servers for Lower Carbon

	University of Washington (CSE), Host: Prof. Baris Kasikci	*Jun 2025*
	MSR India, Host: Dr. Ramachandran Ramjee	*Oct 2024*
	AMD Research, Host: Dr. Srilatha (Bobbie) Manne	*Oct 2024*
	Intel Processor Architecture Research Lab, Host: Dr. Shankar Balachandran	*Oct 2024*
	Princeton University (CS), Host: Prof. Margaret Martonosi	*Oct 2024*
	Rutgers University (CS), Host: Prof. Santosh Nagarakatte	*Oct 2024*
	Systems Research @ Google (SRG) Group, Host: Dr. David Culler	*Sep 2024*
	ISCA, Buenos Aires	*Jun 2024*
	University of California San Diego (CSE), Host: Prof. Jishen Zhao	*Apr 2024*
	University of California Riverside (ECE), Host: Prof. Shaolei Ren	*Apr 2024*
	University of Southern California (ECE), Host: Prof. Murali Annavaram	*Apr 2024*

SELECTED PRESS

– *IEEE Spectrum* *Nov 2024*
"Servers Get a Second Life for Sustainability"

– *MSN* *Oct 2024*
"Researchers discover new method to curb cloud computing's harmful impact: 'We targeted planned obsolescence"'

– *TechXplore* *Oct 2024*
"New methodology enables design of cloud servers for lower carbon"

– *Carnegie Mellon University News* *Aug 2024*
"Enabling Sustainable Cloud Computing with Low-Carbon Server Design"

– *Carnegie Mellon University News* *May 2024*
"Designing Cloud Servers for Lower Carbon Emissions"

AWARDED GRANT PROPOSALS COLLABORATED ON

– *AWS Cloud Credit for Research*, "Carbon-Aware Scheduling to Reduce Hyperscale Carbon Emissions", *Award*: $76,000, Award period: 2024–25

PROFESSIONAL SERVICE

Program committee chair

– *Co-Chair* for Wild and Crazy Ideas *Mar 2026*
(**WACI 2026**) held in conjunction with ASPLOS

Organizer/founder of workshop or panel

– *Workshop Lead Organizer* for 2nd Workshop on Ethical Systems and Architecture Design *Mar 2026*
(**HotEthics 2026**) held in conjunction with ASPLOS

– *Workshop Co-Founder* for 1st Workshop on Ethical Systems and Architecture Design *Apr 2024*
(**HotEthics 2024**) held in conjunction with ASPLOS

– *Panel Co-Organizer* for Panel on Sustainable Systems at The 19th Workshop on *Feb 2023*
Hot Topics in Operating Systems (**HotOS 2023**)

Invited contributor, speaker, or panelist

– *Invited Speaker* for ACT Tutorial (Architectural Carbon Modeling Tool) *Nov 2024*
held in conjunction with MICRO

– *Graduate Student Panelist* for Undergrad Architecture Mentoring Workshop (**uArch**) *Jun 2024*
held in conjunction with ISCA

– *Co-Author* in ACM SIGARCH Computer Architecture Today blog article *Reducing Embodied Carbon is Important* (**ACM SIGARCH 2023**) *Aug 2023*

Artifact evaluation committee member

– Architectural Support for Programming Languages and Operating Systems (**ASPLOS 2025**) *Feb 2025*

– International Symposium on Microarchitecture (**MICRO 2024**) *Sep 2024*

– Architectural Support for Programming Languages and Operating Systems (**ASPLOS 2024**) *Oct 2023*

PROFESSIONAL EXPERIENCE

PhD Software Engineering Intern, Google, Durham, NC *May 2026 – Aug 2026*
continued part-time as **Student Researcher** *Aug 2026 – Dec 2026*
Mentors: Jixin Feng and Soheil Khadir

Leading a project on reducing the power consumption of host servers that support large-scale ML training workloads.

Azure Systems Research Intern, Microsoft Research, Redmond, WA *May 2025 – Aug 2025*
Mentors: Fiodar Kazhamiaka and Daniel Berger

Led research project on novel approaches to optimize data center carbon efficiency through a combination of strategic resource management and workload identification; findings have influenced oversubscription and server-design strategies at Microsoft Azure.

IBM Research Intern, IBM Research, Yorktown Heights, NY *May 2024 – Aug 2024*
Mentor: Asser Tantawi

Created new frameworks and strategies for more sustainable cross-cluster batch job scheduling.

Azure Systems Research Intern, Microsoft Research, Redmond, WA *May 2023 – Aug 2023*
Mentors: Fiodar Kazhamiaka and Daniel Berger

Designed and built a novel framework to understand the data center-scale impacts of server design to improve resource and carbon efficiency. Work has reached early production and is being considered by Azure as a strategy towards achieving Microsoft's 2030 decarbonization targets.

Graduate Research Assistant, Carnegie Mellon University *Aug 2022 – Present*
Advisor: Prof. Akshitha Sriraman

Introducing sustainability as a first-order hardware/software system design metric for hyperscale systems and redesigning data center systems to be carbon-efficient.

Undergraduate Research Assistant, Harvard University *Mar 2021 – Aug 2022*
Lab: Harvard Architecture, Circuits, and Compilers Group
Advisors: Dr. Abdulrahman Mahmoud, Profs. Gu-Yeon Wei and David Brooks

Developed a hardware-aware framework for analyzing the resilience of deep neural network accelerators to soft errors, considering the reuse of values in an accelerator's dataflow propagation.

Engineering Development Group Intern, MathWorks *May 2021 – Aug 2021*
Team: Deep Learning HDL Toolbox
Supervisors: Wang Chen, Siyuan Xu

Enabled efficient mapping of non-square convolution kernels onto square processing-element arrays, allowing users to deploy models using non-square kernels onto FPGAs

Undergraduate Research Assistant, Harvard University *May 2020 – Aug 2020*
Lab: Harvard Edge Computing Lab
Advisor: Prof. Vijay Janapa Reddi

Analyzed how using SLAM for localization affected efficiency and power usage within autonomous drone applications by integrating SLAM algorithms into an open-sourced drone benchmarking framework.

Undergraduate Research Assistant, Harvard University *May 2019 – Aug 2019*
Lab: Hoffman Physics Lab
Advisor: Prof. Jenny Hoffman

Developed a tensioning system for an XY-walker system to extend the range of a scanning tunneling microscope used to research the proximity effect of superconductivity.

STUDENT MENTORING

PhD

- Abnash Bassi *2025–Present*
  First author on EMC2 2026 workshop paper

Master's

- Aaron Hsiung (now at TSMC) *2025–2026*
- Josiah Miggiani *2025*
- Melissa Pan (now a PhD student at UC Berkeley) *2023–2024*
  Co-author on SOSP 2024 poster, YArch 2024 paper, and a paper under submission

Undergraduate

- Onyx Mayer *2025*
- Andrew Gao (now at Vooma) *2024*
- Yuxuan Zhu (now at Intuit) *2023*

TEACHING EXPERIENCE

Graduate teaching assistant, Carnegie Mellon University

– *Computer Architecture and Systems (18-742)*; Prof. Phil Gibbons *Spring 2026*

Invited guest lecture on sustainable computing

– *Computer Systems & the Hardware-Software Interface (18-344)* *Fall 2024*
Profs. Akshitha Sriraman & Brandon Lucia

Undergraduate teaching assistant, Harvard University

– *Systems Programming and Machine Organization (CS 61)*; Prof. Eddie Kohler *Fall 2020, 2021*

– *Circuits, Devices, and Transduction (ES 152)*; Profs. Gage Hills & Woodward Yang *Fall 2021*

– *Systems and Control (ES 150)*; Profs. Na Li & Yue Lu *Fall 2021*

– *Introduction to Electrical Engineering (ES 50)*; Profs. Chris Lombardo & Marko Loncar *Spring 2021*

– *Integration, Series and Differential Equations (MATH 1B)*; Dr. Hakim Walker *Fall 2019*

LEADERSHIP & VOLUNTEERING

– CMU Institute of Technology K-12 Outreach, STEM Volunteer *2023–Present*

– President of Harvard Club Tennis *2019–2022*

– President of Harvard College Engineering Society *2020–2021*

– Co-President of Harvard Engineering Peer Concentration Advisors *2020–2021*

REFERENCES

1. Prof. Akshitha Sriraman (akshitha@cmu.edu)
   Assistant Professor, Carnegie Mellon University
2. Dr. Ricardo Bianchini (ricardob@microsoft.com)
   Technical Fellow, Corporate Vice President, Microsoft Azure
3. Dr. Parthasarathy Ranganathan (parthas@google.com)
   Vice President and Technical Fellow, Google
4. Prof. Babak Falsafi (babak.falsafi@epfl.ch)
   Professor, École Polytechnique Fédérale de Lausanne (EPFL)
5. Prof. David Brooks (dbrooks@g.harvard.edu)
   Haley Family Professor of Computer Science, Harvard University
6. Prof. Udit Gupta (ugupta@cornell.edu)
   Assistant Professor, Cornell University